

BIOLOGY



WHEN YOU LOOK at a running horse, you know immediately that it is alive; a beach pebble, by contrast, is non-living. What distinguishes the two is life, or the state of being alive. Biology is the study of life and living things, and it can be divided into two main fields: zoology and botany. People who study biology are known as biologists; the living organisms they study range from animals such as horses to micro-organisms such as green algae. All use energy obtained from food and released by respiration in order to fulfil their natural processes.

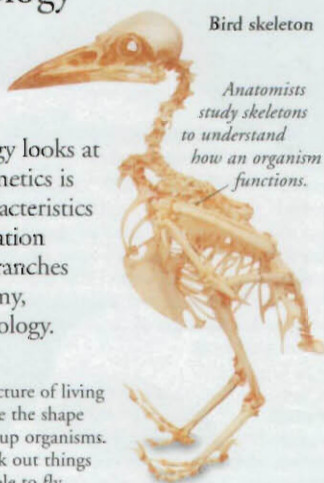
Branches of biology

Biology covers a number of different studies.

Ecology examines how living things interact and where they live. Physiology looks at how organisms work. Genetics is concerned with how characteristics inherited from one generation pass to the next. Other branches include anatomy, taxonomy, microbiology, and parasitology.

Anatomy

Anatomy is the study of the structure of living organisms. Anatomists investigate the shape and form of the parts that make up organisms. This analysis allows them to work out things such as how bats and birds are able to fly.



Bird skeleton

Anatomists study skeletons to understand how an organism functions.



Case displays butterflies and moths

Taxonomy

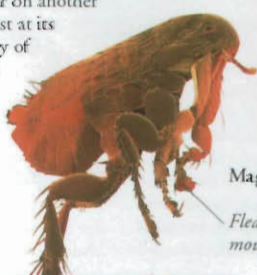
The science of classifying the millions of living things into groups of related organisms is called taxonomy. Scientists called taxonomists identify and name organisms, and then group them together according to the characteristics they share and their common ancestry.

Microbiology

Micro-organisms are living things that are too small to be seen without a microscope. Microbiology is the study of all aspects of the biology of these tiny organisms, which include bacteria, viruses, protists, and some types of fungi such as yeasts.

Parasitology

Parasites live in or on another organism and exist at its expense; the study of parasites is called parasitology. Fleas are parasites that suck blood from their host. Tapeworms live and feed in their host's intestine.



Compound microscope



Magnified flea image

Flea uses needle-like mouth part to suck blood.

Zoology

Zoology is the branch of biology that is concerned with the study of animals. Animals are an amazingly diverse group of living organisms and encompass everything from sponges, spiders, and earthworms to lobsters, cats, and chimpanzees. Zoologists study the structure of animals, how their bodies function, and how they live and behave in their natural environment.



Ethology

The study of animal behaviour is called ethology. Austrian zoologist Konrad Lorenz (1903–89) helped establish the science of ethology. He discovered imprinting, a rapid learning process that occurs early in life. Imprinting to food, surroundings, or mother, happens instinctively during a short, fixed timespan early in life.

Botany

Botany is the study of plants. Plants are diverse organisms, encompassing everything from mosses and ferns to trees, cacti, and flowers. They make their own food by a process called photosynthesis which transforms sunlight into energy. Botanists are concerned with all aspects of the structure, function, and ecology of plants.



Kew Gardens, London, England

Work of a biologist

Biologists are trained in all branches of biology, but usually focus on one specific area. Their research might involve observing animal behaviour, investigating plant photosynthesis, or studying ecosystems.



Petri dishes contain control samples.

Biologist at work in a laboratory

Rachel Carson

In 1962, the American marine biologist and writer Rachel Carson (1907–64) published a book called *The Silent Spring*. In it, she warned that the indiscriminate use of pesticides and weedkillers was poisoning the natural world. Her pioneering book was fundamental in starting the environmental movement and in making ecological information accessible to the public.



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